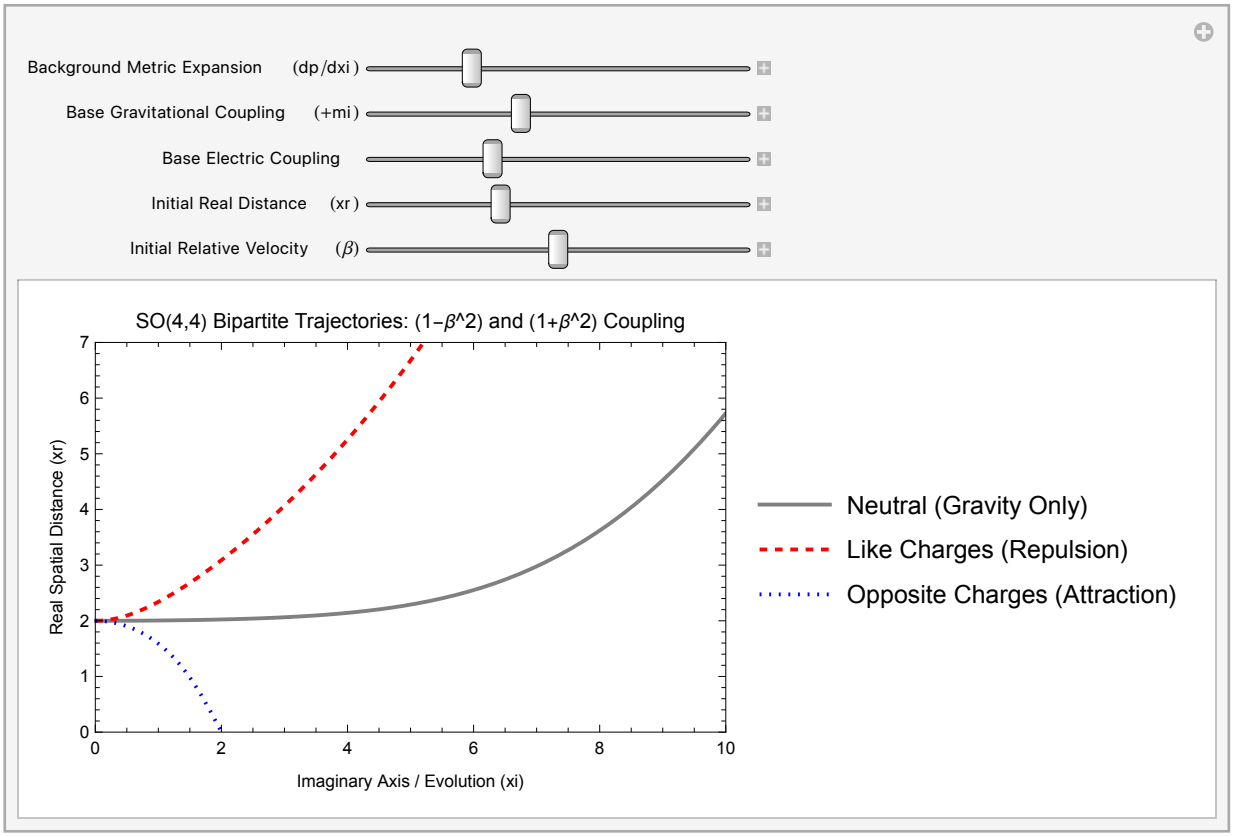
In[•]:=
```
(*phase_kinetics.m:SO(4,4) Feynman Coordinate Trace Sub-routine*)Manipulate[
 Module[{tMax = 10, beta, eqOfMotion, solNeutral, solLike, solOpposite, x, xi, t},
  (*Normalized velocity:beta=v/c.Using c=1 for the coordinate trace*)
  beta[x_, xi_] := x'[xi];
  (*Equation of Motion in the xr (Real) plane.Acceleration=
   (Background Metric Expansion)-(Gravity*(1+beta^2))+
    (Electric*ChargeSign*(1-beta^2))*)eqOfMotion[chargeSign_] :=
   x''[xi] == metricPressure - (gForce * (1 + beta[x, xi]^2) / x[xi]^2) +
     (chargeSign * eForce * (1 - beta[x, xi]^2) / x[xi]^2);
  (*1. Neutral Particles (Gravity Only,ChargeSign=0)*)
  solNeutral = Quiet@NDSolveValue[
     {eqOfMotion[0], x[0] == initialXr, x'[0] == initialBeta}, x, {xi, 0, tMax}];
  (*2. Like Charges (Repulsive,ChargeSign=+1)*)solLike = Quiet@NDSolveValue[
     {eqOfMotion[1], x[0] == initialXr, x'[0] == initialBeta}, x, {xi, 0, tMax}];
  (*3. Opposite Charges (Attractive,ChargeSign=-1)
    Includes a WhenEvent to terminate the trace exactly when xr→
   0 (The 2D superposition boundary collapse)*)solOpposite =
   Quiet@NDSolveValue[{eqOfMotion[-1], x[0] == initialXr, x'[0] == initialBeta,
      WhenEvent[x[xi] ≤ 0.01, "StopIntegration"]}, x, {xi, 0, tMax}];
  (*Plotting the Feynman-style coordinate traces*)
  Plot[{solNeutral[xi], solLike[xi], solOpposite[xi]},
   {xi, 0, tMax}, PlotStyle → {Directive[Thick, Gray],
     Directive[Thick, Red, Dashed], Directive[Thick, Blue, Dotted]},
   PlotRange → {{0, tMax}, {0, initialXr + 5}}, AxesOrigin → {0, 0}, Frame → True,
   FrameLabel → {"Imaginary Axis / Evolution (xi)", "Real Spatial Distance (xr)"},
   PlotLegends → {"Neutral (Gravity Only)",
     "Like Charges (Repulsion)", "Opposite Charges (Attraction)"},
   PlotLabel → "SO(4,4) Bipartite Trajectories: (1-β^2) and (1+β^2) Coupling"]],
 (*Simulation Sliders*){{metricPressure, 0.5,
   "Background Metric Expansion (dp/dxi)"}, 0, 2},
 {{gForce, 1.0, "Base Gravitational Coupling (+mi)"}, 0, 5},
 {{eForce, 3.0, "Base Electric Coupling"}, 0, 10},
 {{initialXr, 2.0, "Initial Real Distance (xr)"}, 0.5, 5},
 {{initialBeta, 0.0, "Initial Relative Velocity (β)"}, -0.9, 0.9}]
```

Out[ ]=


Background Metric Expansion (dp/dxi)
Base Gravitational Coupling (+mi)
Base Electric Coupling
Initial Real Distance (xr)
Initial Relative Velocity (β)
SO(4,4) Bipartite Trajectories: (1−β^2) and (1+β^2) Coupling
Real Spatial Distance (xr)
Imaginary Axis / Evolution (xi)
0
1
2
3
4
5
6
7
0
2
4
6
8
10
Neutral (Gravity Only)
Like Charges (Repulsion)
Opposite Charges (Attraction)